

Quantifying Cognitive Fatigue in Real Time Using EEG-Derived Engagement Metrics (3207)

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Abstract

Many students rely on prolonged study sessions and cramming, despite evidence that sustained cognitive effort leads to fatigue and reduced performance. This study investigates whether cognitive fatigue can be detected using a consumer-grade EEG system during academic tasks. EEG data were collected from six participants using a Muse 2 headband while completing sustained cognitive activities. Frequency band powers were extracted and used to compute an Engagement Index (EI), representing cognitive engagement. Across participants, EI showed consistent declines over time, indicating reduced engagement during prolonged tasks. These patterns were reproducible, suggesting the signal reflects genuine cognitive fatigue rather than random variability. Overall, the results demonstrate that consumer EEG can detect meaningful changes in cognitive state, supporting its potential for real-time study optimization and future neurorehabilitation applications.

Introduction

Students are often encouraged to take breaks using fixed schedules such as the Pomodoro technique [1]. However, attention is not constant — both individuals and cognitive demands vary over time. Fixed intervals may occur too late, after fatigue has already impaired performance, or too early, interrupting productive focus. To optimize learning, break timing must adapt to the individual. The brain undergoes continuous reorganization, a property known as neuroplasticity, and cognitive performance depends on maintaining an optimal load. When mental effort exceeds this threshold, performance declines [2]. EEG allows real-time monitoring of neural activity to detect early signs of cognitive fatigue before observable performance drops, enabling adaptive, personalized study-rest cycles based on neural data rather than fixed intervals [3].

Figure 1. EEG measures electrical potentials generated by synchronized postsynaptic activity of cortical pyramidal neurons. Surface electrodes detect voltage fluctuations that can be decomposed into frequency components using spectral analysis methods such as Fast Fourier Transform (FFT). Power spectral density (PSD) analysis quantifies neural oscillations within canonical frequency bands—delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (13–30 Hz), and gamma (30+ Hz)—each associated with distinct cognitive processes [4].

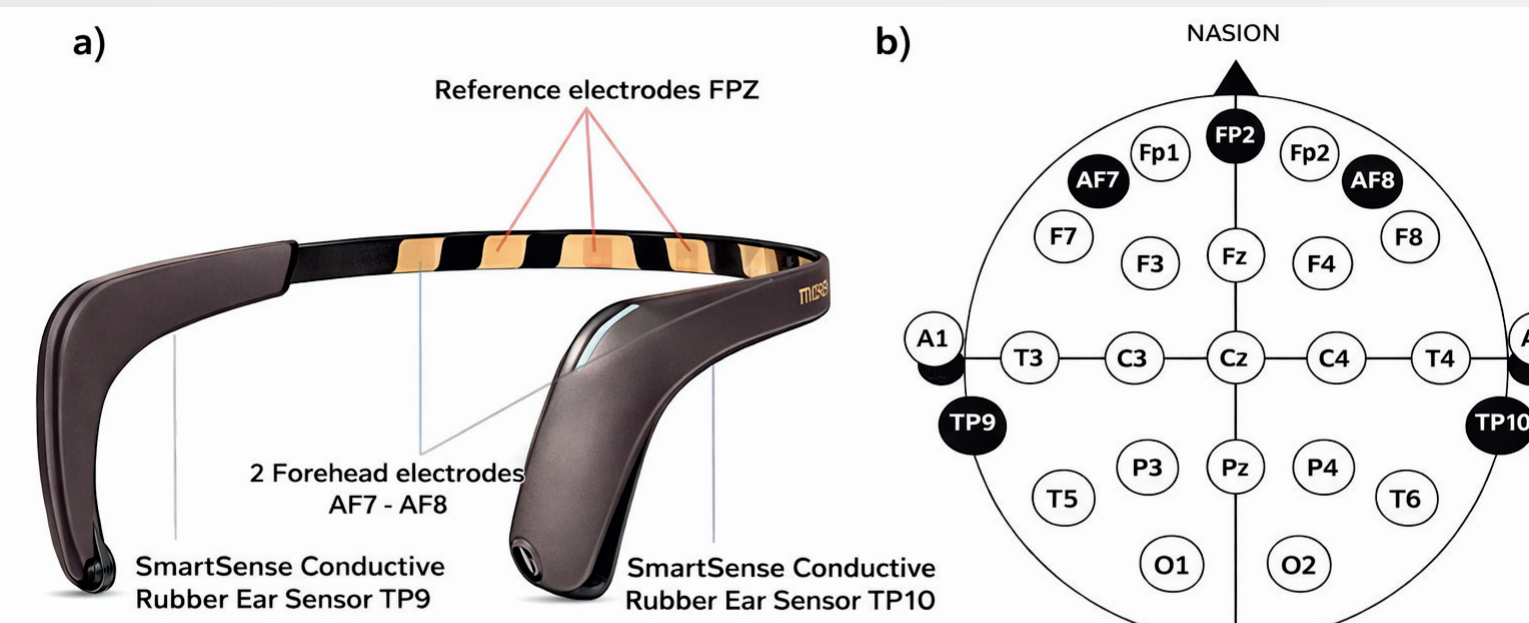
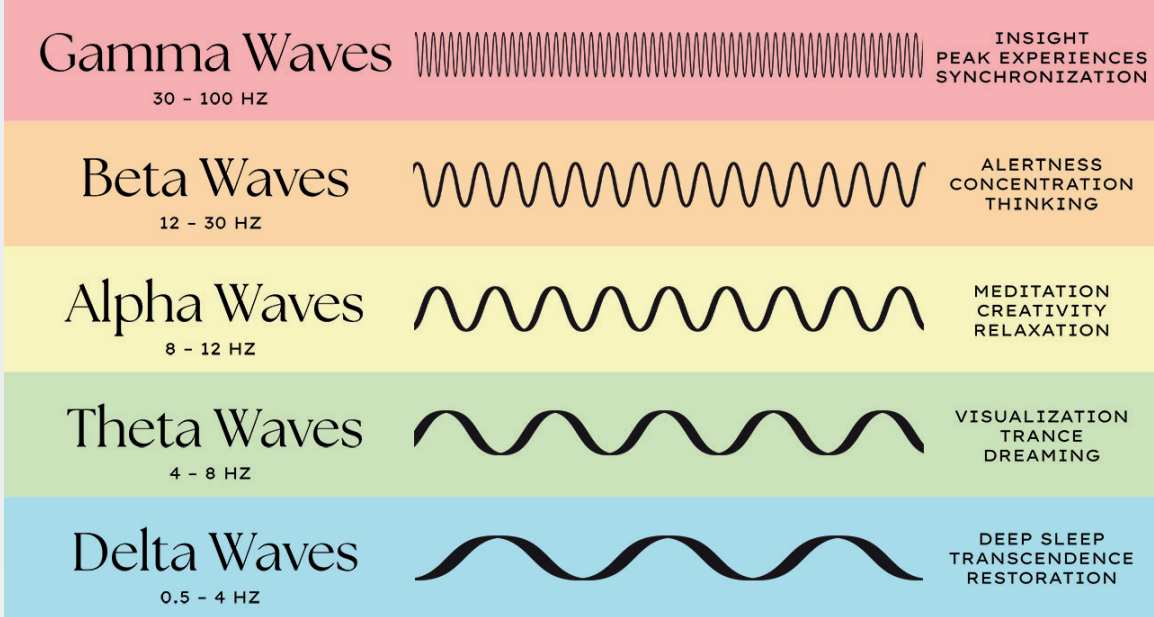


Figure 2. The Muse 2, the consumer grade EEG used in this study. The headband uses 4 electrodes (AF7,AF8,TP9,TP10)and 3 reference electrodes (Fp1,Fp2,Fp3)to measure activity in the frontal and parietal lobes [5].

$$EI(t) = \frac{\beta(t)}{\alpha(t) + \theta(t)}$$

Figure 3. As cognitive fatigue develops, characteristic changes occur in EEG frequency bands. Beta activity, associated with active concentration, decreases over time, while alpha and theta activity increase, reflecting reduced alertness and growing fatigue. These opposing trends can be combined into an engagement metric, defined as the ratio of beta power to the sum of alpha and theta power [6].This Engagement Index provides a quantitative measure of cognitive engagement and enables the detection of fatigue based on neural activity rather than behavioral observation.

Research Question:

Can cognitive fatigue be quantified in real time using EEG-derived engagement metrics from a consumer-grade wearable device?

Hypothesis:

As cognitive fatigue increases, beta activity will decrease while alpha and theta activity increase, resulting in a measurable decline in the engagement index over time.

Methodology

Participants:

- n = 6 students (ages 16 ± 1 years)
- All participants were healthy with no self-reported neurological conditions or medication affecting cognitive performance. Informed consent obtained in accordance with Youth Science Canada (YSC) ethics guidelines

Experimental Procedure:

- Participants were seated in a quiet, well-ventilated environment
- Muse 2 headband was fitted according to protocol, ensuring stable signal quality
- Participants remained still and avoided speaking to reduce motion artifacts
- EEG was recorded continuously for 30 minutes while the researcher remained outside the room
- Participants worked on an SAT Mocks Exam during the recording period
- Tasks were selected to induce sustained cognitive load and promote mental fatigue
- After recording, the device was removed and participants completed a self-report questionnaire assessing focus, fatigue, task difficulty, and discomfort

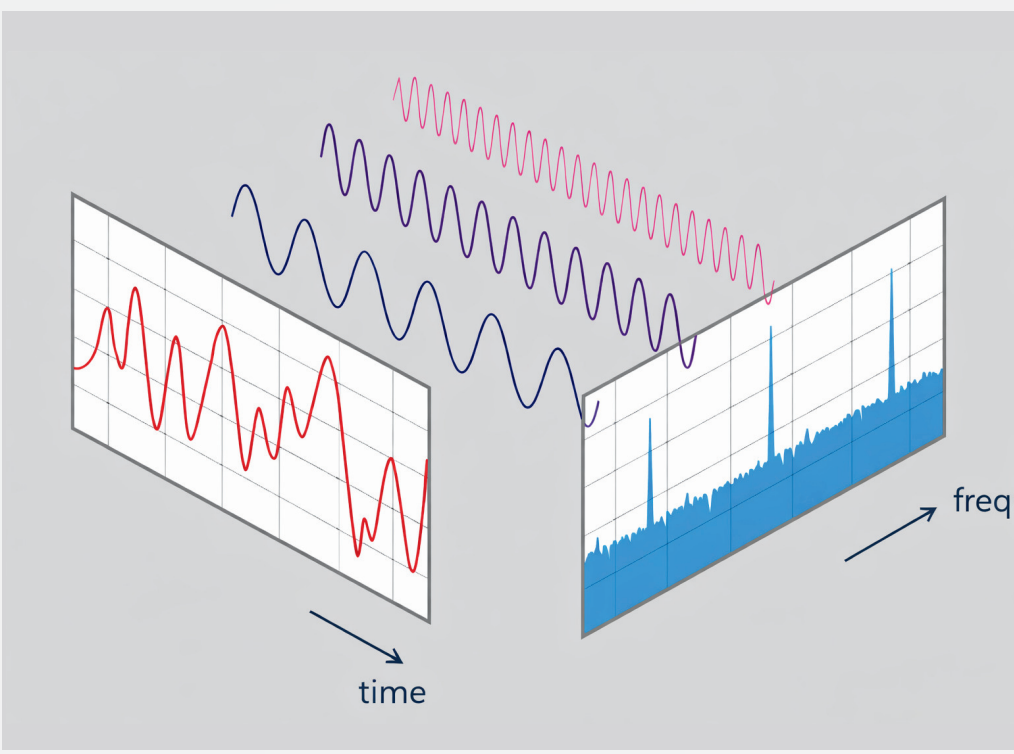
High Level EEG Data Processing Pipeline:

Obtain Raw EEG Data

EEG data was collected using the Muse 2 headband and exported via Muse Monitor as CSV files. The dataset includes raw time-domain signals.

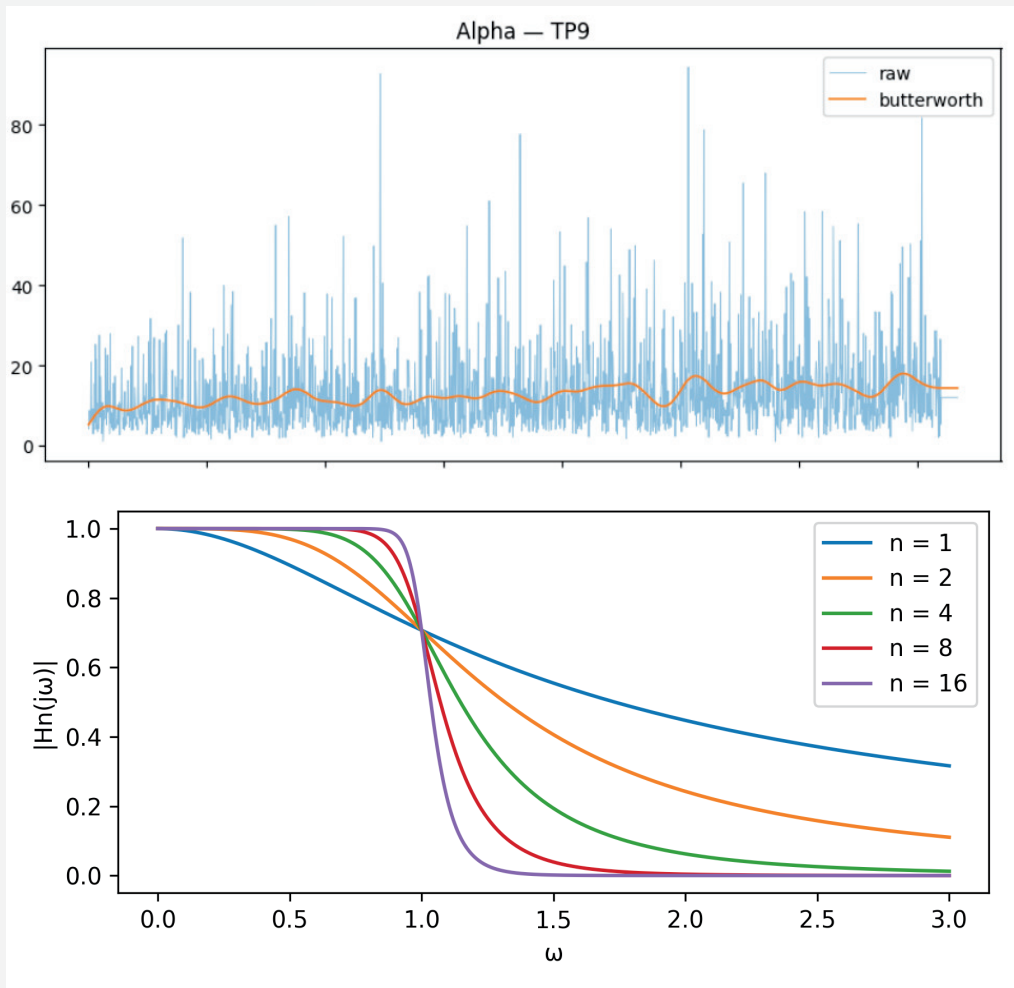
Fast Fourier Transform

Figure 4. The raw time-domain EEG signals were transformed into the frequency domain using the Fast Fourier Transform (FFT). This process decomposes the raw signal from each of the 4 electrodes into their 5 frequency components [7].This results in 20, each representing the power of a specific frequency band over time.



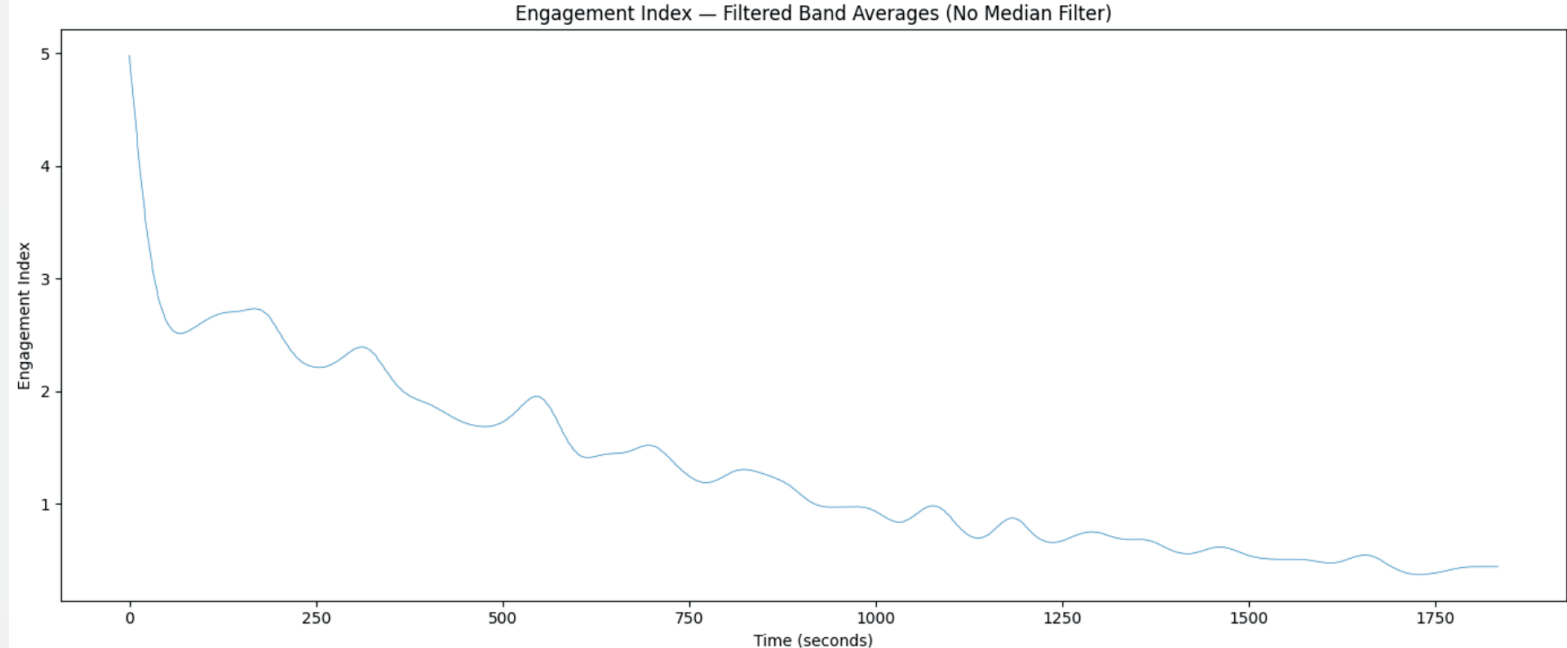
Apply Low-Pass Butterworth Filter

Figure 5. A low-pass Butterworth filter was applied to each of the 20 EEG signals to reduce high-frequency noise and artifacts. As shown in the top panel, the filtered signal (orange) smooths rapid spikes in the raw signal (blue), removing noise from sources such as facial muscle activity and headband movement. The filter is defined by two parameters: cutoff frequency, which sets where higher frequencies begin to be reduced, and order, which controls how sharply this transition occurs, as illustrated in the bottom panel [8].



Construct Engagement Index

Figure 6. After averaging across electrodes, we obtain 5 band power time series, one for each frequency band, representing how power changes over time. The Engagement Index (EI), described in Fig. 3, is then computed using the alpha, beta, and theta band power signals. This results in a single Engagement Index time series that reflects cognitive engagement over time.



EEG Fatigue Detection Signal

A 2–10 minute baseline was used to calculate the mean and standard deviation of the Engagement Index (EI). A threshold of mean − 1.5 SD was applied, with values below indicating fatigue. This allows detection to be personalized and adjusted based on user calibration. The final output is a time series of Engagement Index values. In the live model, this pipeline continues looping for every new data point, allowing us to give the user live stimuli in the form of audio.

Results

Figure 7. Correlation heat map of average EEG band powers from representative sample in the data set. Notice how the Alpha and Theta bands have a positive correlation whilst a negative one with the Beta frequency band.

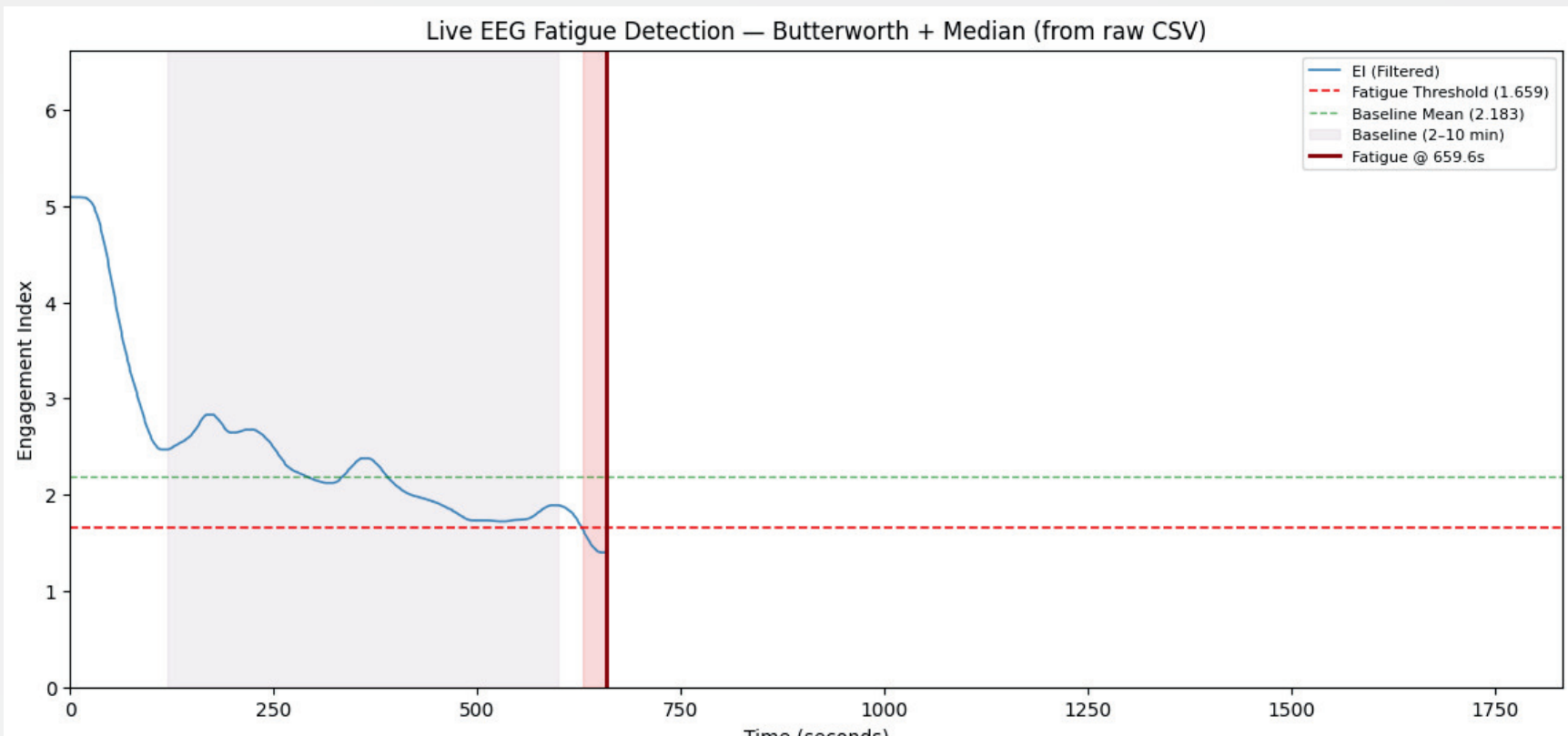
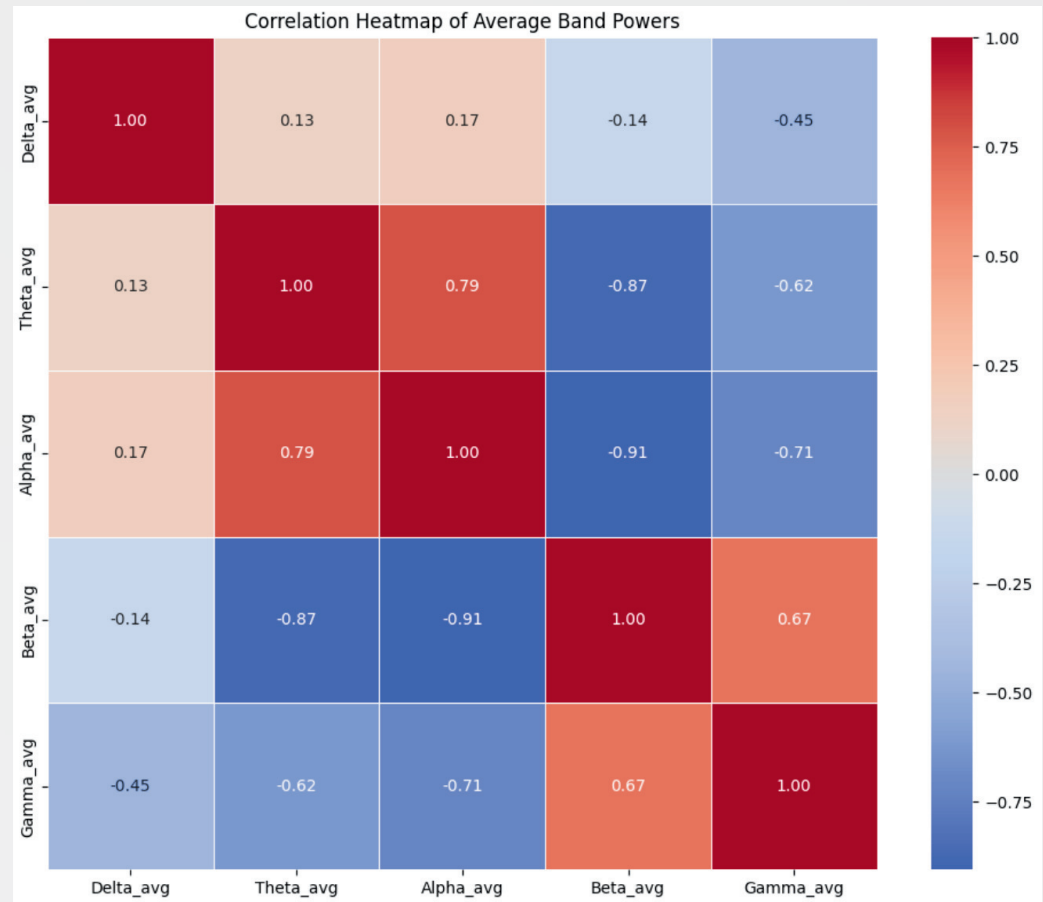


Figure 8. Engagement Index (EI) over time for Participant A. The green dashed line shows the baseline mean, the red dashed line shows the fatigue threshold (mean − 1.5 SD), and shaded regions indicate baseline and fatigue periods.

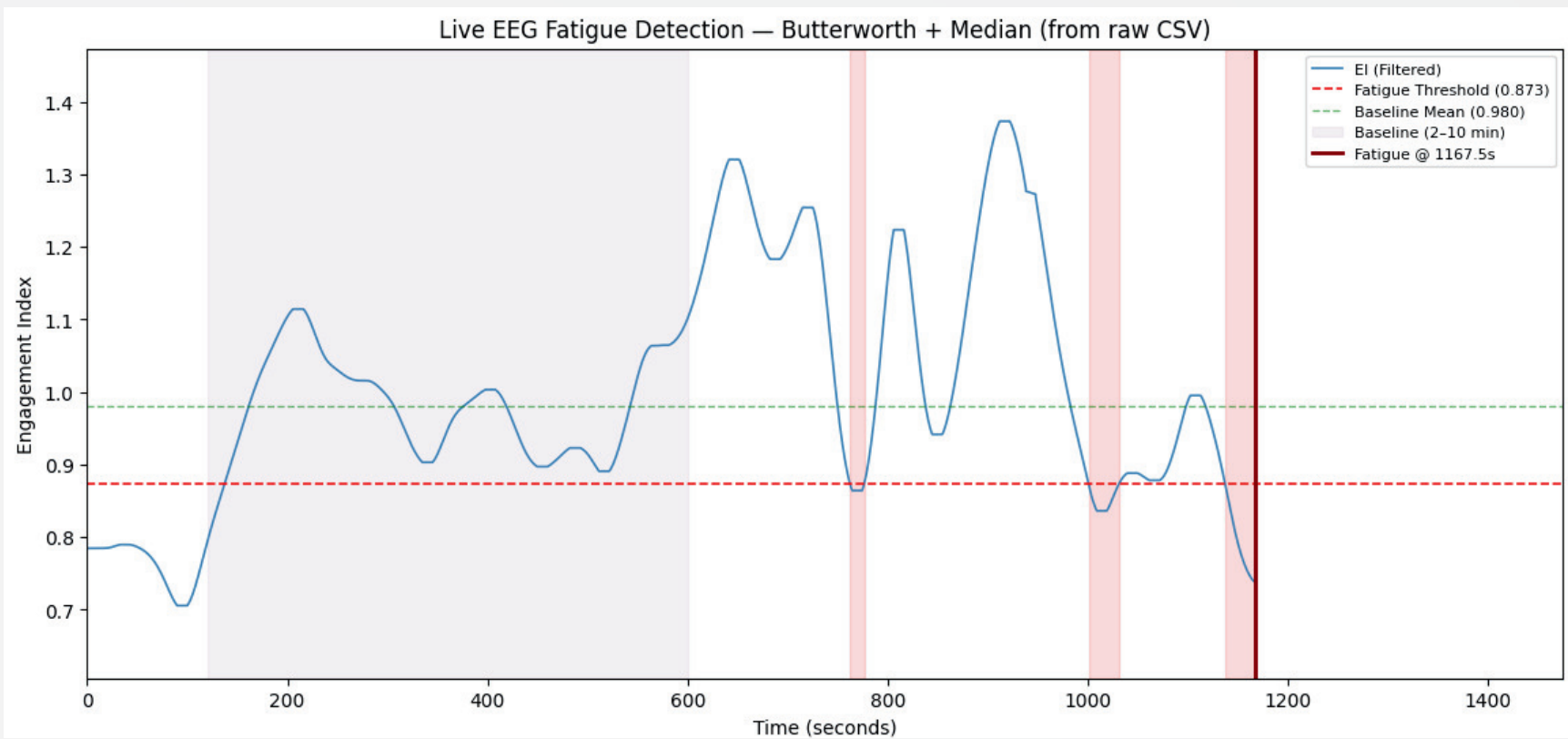


Figure 9. Engagement Index (EI) over time for Participant B

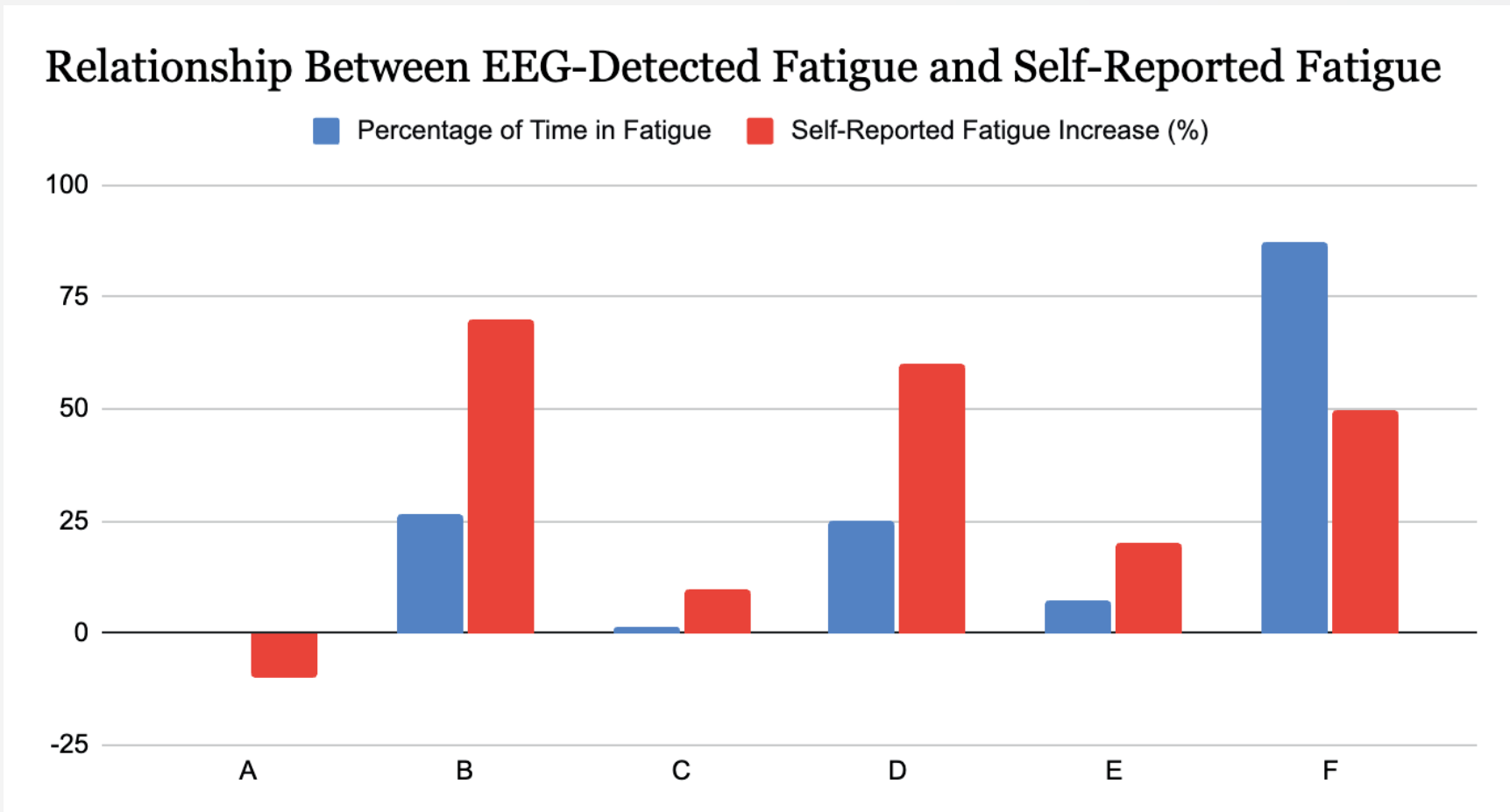


Figure 10. Comparison of EEG-detected fatigue (percentage of time below threshold) and self-reported fatigue increase across participants. Overall, higher EEG-detected fatigue generally corresponds with greater self-reported fatigue.

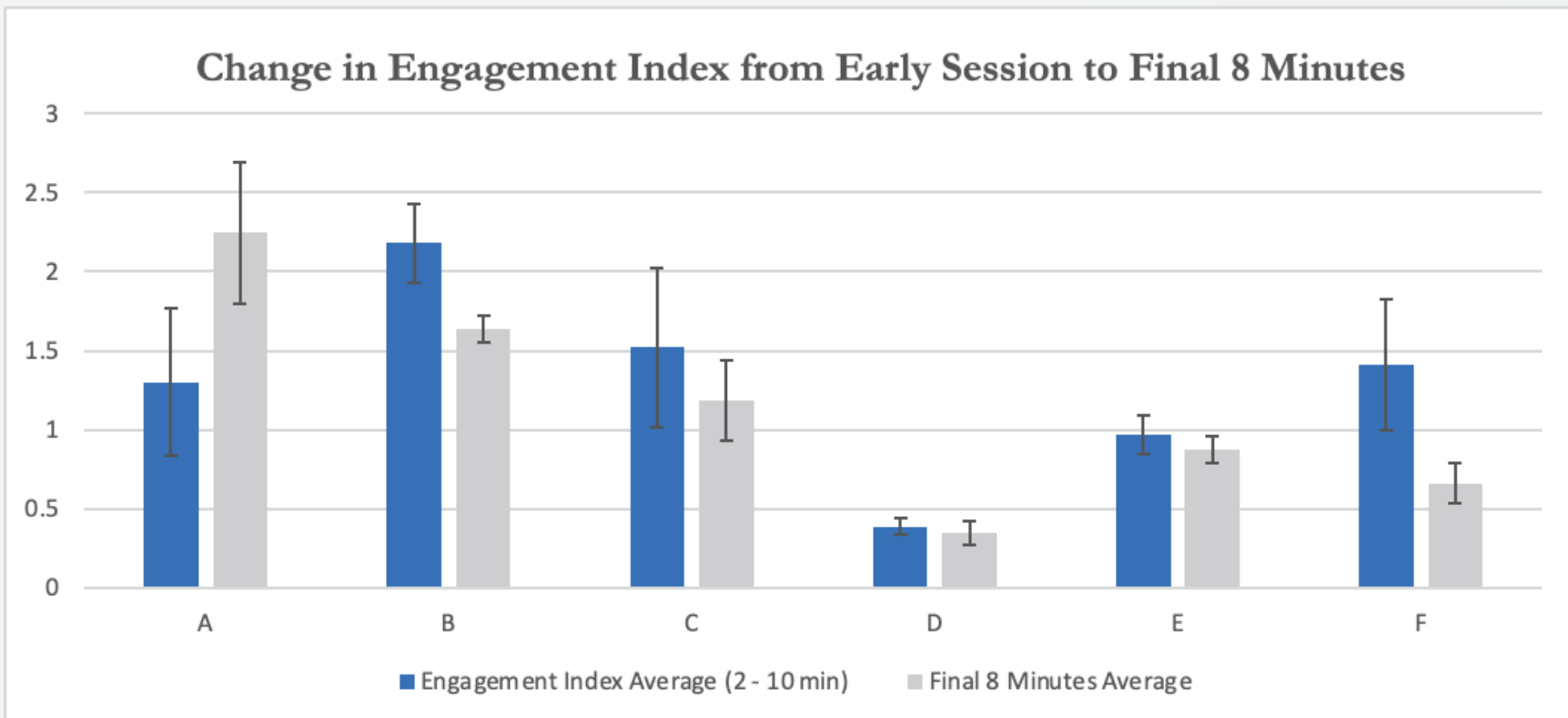


Figure 11. Comparison of mean Engagement Index between the early session window (2–10 minutes) and the final 8 minutes across participants (A–F). Error bars represent 95% confidence intervals.

Discussion

The results support the hypothesis that cognitive fatigue can be detected using EEG-derived engagement metrics. Across participants, the Engagement Index (EI) generally decreased over time during sustained cognitive tasks (Figures 8 and 9), with multiple instances falling below the fatigue threshold. These threshold crossings indicate periods of reduced engagement consistent with cognitive fatigue.

This trend is further supported by the comparison between early-session and final-session EI values (Figure 11), where most participants show a decrease in mean EI. Despite individual variability, the overall pattern suggests that prolonged cognitive effort leads to measurable declines in neural engagement.The relationship between EEG-detected fatigue and self-reported fatigue (Figure 10) provides additional validation. Participants who spent a greater percentage of time below the fatigue threshold generally reported higher fatigue, indicating that the Engagement Index reflects meaningful changes in subjective cognitive state.The correlation heatmap (Figure 7) aligns with the theoretical basis of the metric, showing strong positive relationships between alpha and theta bands and negative correlations with beta activity. This supports the use of the Engagement Index formulation as a proxy for cognitive engagement.

Overall, these findings demonstrate that a consumer-grade EEG device can detect consistent patterns of cognitive fatigue. While variability and a small sample size (n = 6) limit generalizability, the results show strong potential for real-time fatigue monitoring and adaptive learning systems.

Conclusion

The results support the hypothesis that cognitive fatigue can be detected using EEG-derived engagement metrics, as the Engagement Index consistently decreased over time and aligned with self-reported fatigue across participants. This demonstrates that even consumer-grade EEG devices—not just lab-grade systems—have the capability to detect meaningful changes in cognitive state, answering the research question and supporting real-time fatigue monitoring applications.

Figure 12. Visual Differences between a research grade (A) and consumer grade (B) EEG based BCI.



Applications and Future Steps

Beyond Education

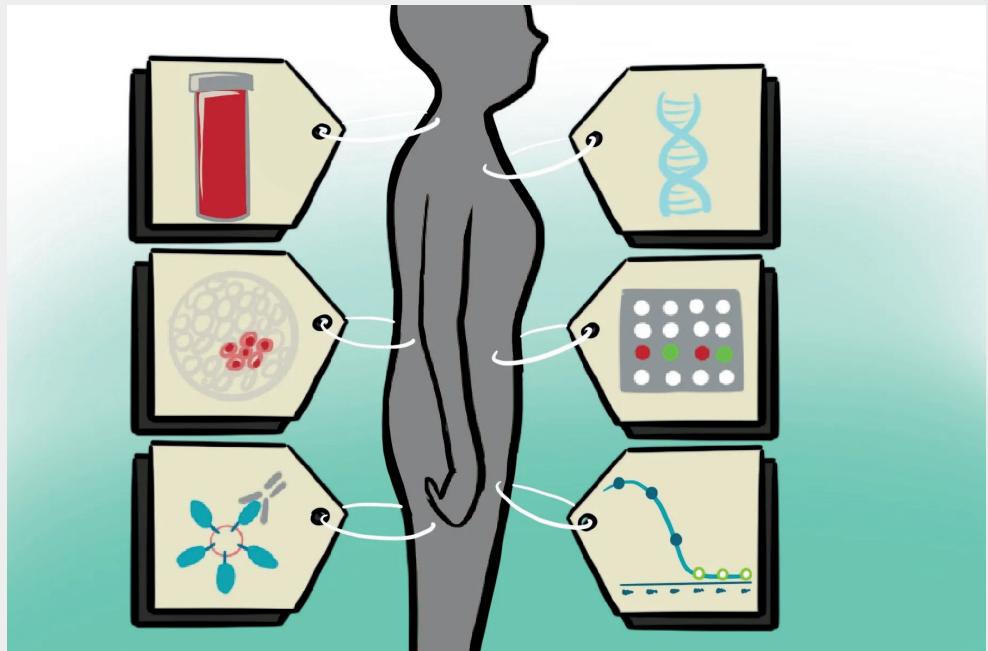
- EEG-based fatigue monitoring has potential clinical applications. Similar systems could support stroke rehabilitation by reinforcing beneficial neural activity patterns through neuroplasticity.They may also assist in monitoring attention disorders, tracking cognitive fatigue during concussion recovery, and supporting cognitive therapy.

Personalized Fatigue Prediction

- Future work will integrate machine learning to learn individual fatigue patterns and predict optimal, personalized break timing using past session data

Biomarker Validation

- Figure 13.** The collected EEG data can be compared with blink rate and heart rate as well as other biomarkers to further strengthen the validity of the Engagement Index as a biomarker of cognitive fatigue. By analyzing patterns across multiple, we can further our understanding of how cognitive overload develops over time.



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